

Averages & Weighted Averages

Topic Test — Solutions & Answers (Part 2: Q21–Q32)

Question 21

A batsman played $n + 2$ innings and got out on all occasions. His average score in these $n + 2$ innings was 29 runs and he scored 38 and 15 runs in the last two innings. The batsman scored less than 38 runs in each of the first n innings. In these n innings, his average score was 30 runs and lowest score was x runs. The smallest possible value of x is

- A) 4
- B) 3
- ✓ C) 2 ← Correct Answer
- D) 1

Correct Answer: C

Solution:

Step 1: Find the value of n

Total runs in $n+2$ innings = Average $\times$ Number of innings = $29(n+2)$.

Runs in the last two innings = $38 + 15 = 53$.

Total runs in the first n innings = $29(n+2) - 53 = 29n + 58 - 53 = 29n + 5$.

The average score in the first n innings is given as 30, so total runs = $30n$.

Equating: $30n = 29n + 5 \rightarrow n = 5$.

Step 2: Set up the problem to find the minimum score

The batsman played 5 innings. Total score in these 5 innings = $30 \times 5 = 150$.

Each of these scores is less than 38, so each score ≤ 37 .

Let the lowest score be $x = s_1$. To find the smallest possible x , maximize the other four scores.

Step 3: Calculate the smallest possible value of x

Set s_2, s_3, s_4, s_5 to their maximum possible value of 37:

$s_1 + 37 + 37 + 37 + 37 = 150 \rightarrow s_1 + 148 = 150 \rightarrow s_1 = 2$.

The smallest possible value for x is 2.

Question 22

The average height of 22 toddlers increases by 2 inches when two of them leave this group. If the average height of these two toddlers is one-third the average height of the original 22, then the average height, in inches, of the remaining 20 toddlers is

- A) 30
- B) 28
- ✓ C) 32 ← Correct Answer
- D) 26

Correct Answer: C

Solution:

Relate the sums of heights:

$$22 \times H_{\text{orig}} = (2/3)H_{\text{orig}} + 20 \times (H_{\text{orig}} + 2)$$

$$22 H_{\text{orig}} = (2/3)H_{\text{orig}} + 20 H_{\text{orig}} + 40$$

$$22 H_{\text{orig}} - 20 H_{\text{orig}} - (2/3)H_{\text{orig}} = 40$$

$$2 H_{\text{orig}} - (2/3)H_{\text{orig}} = 40 \rightarrow (4/3)H_{\text{orig}} = 40 \rightarrow H_{\text{orig}} = 30 \text{ inches.}$$

Question 23

Dick is thrice as old as Tom and Harry is twice as old as Dick. If Dick's age is 1 year less than the average age of all three, then Harry's age, in years, is

[Type in the Answer] Correct Answer: 18

Solution:

Step 1: Set up equations based on the given information

Let Tom's age be T, Dick's age be D, and Harry's age be H.

$$D = 3T \text{ and } H = 2D = 2(3T) = 6T.$$

Step 2: Express the average age in terms of T

$$\text{Average age of the three} = (T + D + H) / 3 = (T + 3T + 6T) / 3 = 10T / 3.$$

Step 3: Use the condition relating Dick's age to the average age

$$\text{Dick's age is 1 year less than the average age: } D = (\text{Average Age}) - 1.$$

$$3T = (10T / 3) - 1.$$

Step 4: Solve the equation for T

$$\text{Multiply by 3: } 9T = 10T - 3 \rightarrow 3 = 10T - 9T \rightarrow T = 3.$$

Step 5: Calculate Harry's age

$$H = 6T = 6 \times 3 = 18. \text{ Harry is 18 years old.}$$

Question 24

In a group of 10 students, the mean of the lowest 9 scores is 42 while the mean of the highest 9 scores is 47. For the entire group of 10 students, the maximum possible mean exceeds the minimum possible mean by

A) 5

✓ B) 4 ← **Correct Answer**

C) 3

D) 6

Correct Answer: B

Solution:

1. Formulate equations from the given means.

Let the 10 scores in non-decreasing order be $s_1, s_2, \dots, s_{10}$.

Sum of the lowest 9 scores: $s_1 + s_2 + \dots + s_9 = 9 \times 42 = 378$ (1)

Sum of the highest 9 scores: $s_2 + s_3 + \dots + s_{10} = 9 \times 47 = 423$ (2)

2. Find $s_{10} - s_1$.

Subtract (1) from (2): $s_{10} - s_1 = 423 - 378 = 45$.

3. Express the total mean and find its minimum value.

Total sum $S = s_1 + (s_2 + \dots + s_{10}) = s_1 + 423$.

Mean $M = S/10 = (s_1 + 423) / 10$. To minimize M , minimize s_1 .

From ordering, $s_1 \leq s_2$ and also $s_1 \leq \text{mean of highest 9} = 47$. Also $s_1 \leq s_2$.

Using $s_1 = s_{10} - 45$, we get $s_1 \geq 47 - 45 = 2$. Minimum $s_1 = 2$.

Minimum mean $M_{\min} = (2 + 423) / 10 = 42.5$.

4. Find the maximum value of the mean.

To maximize M , maximize s_1 . From ordering, $s_1 \leq \text{mean of lowest 9} = 42$.

Maximum $s_1 = 42$. Maximum mean $M_{\max} = (42 + 423) / 10 = 46.5$.

5. Calculate the difference.

Difference = $M_{\max} - M_{\min} = 46.5 - 42.5 = 4$.

Question 25

There are four numbers such that average of first two numbers is 1 more than the first number, average of first three numbers is 2 more than average of first two numbers, and average of first four numbers is 3 more than average of first three numbers. Then, the difference between the largest and the smallest numbers, is

[Type in the Answer] Correct Answer: 15

Solution:

1. From the first condition:

Average of first two = 1 more than first number: $(a + b) / 2 = a + 1$.

$$a + b = 2a + 2 \rightarrow b = a + 2.$$

2. From the second condition:

Average of first three = 2 more than average of first two.

$$(a + b + c) / 3 = (a + b) / 2 + 2 = (a + 1) + 2 = a + 3.$$

$$a + b + c = 3a + 9. \text{ Substitute } b = a + 2: a + (a+2) + c = 3a + 9 \rightarrow c = a + 7.$$

3. From the third condition:

Average of first four = 3 more than average of first three.

$$(a + b + c + d) / 4 = (a + b + c) / 3 + 3 = (a + 3) + 3 = a + 6.$$

$$a + b + c + d = 4a + 24. \text{ Substitute } b = a+2, c = a+7:$$

$$a + (a+2) + (a+7) + d = 4a + 24 \rightarrow 3a + 9 + d = 4a + 24 \rightarrow d = a + 15.$$

4. Find the difference:

The four numbers are $a, a+2, a+7, a+15$.

$$\text{Difference} = \text{largest} - \text{smallest} = (a + 15) - a = 15.$$

Question 26

The average of 30 integers is 5. Among these 30 integers, there are exactly 20 which do not exceed 5. What is the highest possible value of the average of these 20 integers?

- A) 3.5
- B) 5
- ✓ C) 4.5 ← Correct Answer
- D) 4

Correct Answer: C

Solution:

1. Calculate the total sum of the 30 integers.

Total Sum = Average $\times$ Number of integers = $5 \times 30 = 150$.

2. Understand the two groups of integers.

Group 1: 20 integers that do not exceed 5 (each integer ≤ 5).

Group 2: The remaining 10 integers must exceed 5 (each integer > 5).

3. Formulate the goal.

To maximize the average of Group 1, we must maximize Sum(Group 1).

Since Sum(Group 1) + Sum(Group 2) = 150, we must minimize Sum(Group 2).

4. Minimize the sum of Group 2.

Group 2 contains 10 integers each strictly greater than 5 (i.e., ≥ 6 as integers).

Minimum Sum(Group 2) = $10 \times 6 = 60$.

5. Calculate the maximum sum and average of Group 1.

Maximum Sum(Group 1) = $150 - 60 = 90$.

Highest Average = $90 / 20 = 4.5$.

Question 27

An elevator has a weight limit of 630 kg. It is carrying a group of people of whom the heaviest weighs 57 kg and the lightest weighs 53 kg. What is the maximum possible number of people in the group?

[Type in the Answer] Correct Answer: 11

Solution:

Step 1: Understand the objective and constraints.

Objective: Maximize the number of people N in the elevator.

Constraint 1: Total weight ≤ 630 kg.

Constraint 2: The group contains at least one person weighing 57 kg and at least one weighing 53 kg.

Step 2: Formulate a strategy.

To maximize N for a fixed total weight, minimize average weight per person.

Apart from the one heaviest person (57 kg), all other $(N-1)$ people have the minimum weight of 53 kg.

Step 3: Set up the inequality.

$$\text{Total Weight} = 57 + (N-1) \times 53 \leq 630.$$

$$57 + 53N - 53 \leq 630 \rightarrow 53N + 4 \leq 630 \rightarrow 53N \leq 626.$$

Step 4: Solve for N.

$N \leq 626 / 53 \approx 11.81$. Since N must be an integer, the maximum is $N = 11$.

Question 28

A class consists of 20 boys and 30 girls. In the mid-semester examination, the average score of the girls was 5 higher than that of the boys. In the final exam, however, the average score of the girls dropped by 3 while the average score of the entire class increased by 2. The increase in the average score of the boys is

✓ **A) 9.5 ← Correct Answer**

B) 10

C) 4.5

D) 6

Correct Answer: A

Solution:

Step 1: Set up variables for mid-semester scores.

Number of boys = 20, Number of girls = 30.

Let the average score of boys in mid-semester be B. Average score of girls = $B + 5$.

Step 2: Calculate the mid-semester class average.

$$\text{Total mid-semester score} = (20 \times B) + 30 \times (B + 5) = 20B + 30B + 150 = 50B + 150.$$

$$\text{Mid-semester class average} = (50B + 150) / 50 = B + 3.$$

Step 3: Set up variables for final exam scores.

The class average increased by 2: Final class average = $(B + 3) + 2 = B + 5$.

The girls' average dropped by 3: Final girls' average = $(B + 5) - 3 = B + 2$.

Let the increase in the boys' average be x. Final boys' average = $B + x$.

Step 4: Formulate an equation for the final total score.

$$\text{Total final score} = (20 \times (B + x)) + (30 \times (B + 2)) = 20B + 20x + 30B + 60 = 50B + 20x + 60.$$

$$\text{Also, Total final score} = 50 \times (B + 5) = 50B + 250.$$

Step 5: Solve for x.

$$50B + 20x + 60 = 50B + 250 \rightarrow 20x = 190 \rightarrow x = 9.5.$$

Question 29

In an apartment complex, the number of people aged 51 years and above is 30 and there are at most 39 people whose ages are below 51 years. The average age of all the people in the apartment complex is 38 years. What is the largest possible average age, in years, of the people whose ages are below 51 years?

A) 27

B) 25

C) 26

✓ D) 28 ← Correct Answer

Correct Answer: D

Solution:

To maximize A2:

Since $A1 \geq 51$, the term $(38 - A1)$ is negative. To maximize A2, minimize the magnitude of this negative term.

1. Minimize $|38 - A1|$: set A1 as close to 38 as possible. Minimum A1 = 51.

2. Maximize N2: maximum N2 = 39.

Substitute A1 = 51 and N2 = 39:

$$A2_{\max} = 38 + 30 \times (38 - 51) / 39 = 38 + 30 \times (-13) / 39 = 38 - 390/39 = 38 - 10 = 28.$$

Question 30

Ramesh and Gautam are among 22 students who write an examination. Ramesh scores 82.5. The average score of the 21 students other than Gautam is 62. The average score of all the 22 students is one more than the average score of the 21 students other than Ramesh. The score of Gautam is

A) 53

✓ B) 51 ← Correct Answer

C) 48

D) 49

Correct Answer: B

Solution:

Relate the averages:

We are given $A22 = A21(-R) + 1$.

$$A22 = \Sigma22 / 22. A21(-R) = (\Sigma22 - S_R) / 21.$$

$$\text{So: } (\Sigma22 / 22) = ((\Sigma22 - S_R) / 21) + 1.$$

Solve for $\Sigma22$:

$$\text{Multiply by } 22 \times 21 = 462: 21\Sigma22 = 22(\Sigma22 - S_R) + 462.$$

$$21\Sigma22 = 22\Sigma22 - 22S_R + 462.$$

$$-\Sigma22 = -22S_R + 462 \rightarrow \Sigma22 = 22S_R - 462.$$

$$\Sigma22 = 22(82.5) - 462 = 1815 - 462 = 1353.$$

Find Gautam's score:

$$\Sigma22 = 1302 + S_G \rightarrow 1353 = 1302 + S_G \rightarrow S_G = 51.$$

Question 31

A CAT aspirant appears for a certain number of tests. His average score increases by 1 if the first 10 tests are not considered, and decreases by 1 if the last 10 tests are not considered. If his average scores for the first 10 and the last 10 tests are 20 and 30, respectively, then the total number of tests taken by him is

[Type in the Answer] Correct Answer: 60

Solution:

Step 1: Formulate an equation for the first scenario

If the first 10 tests are not considered: Sum of first 10 tests = $10 \times 20 = 200$.

New total score $S' = nA - 200$. New number of tests $n' = n - 10$. New average $A' = A + 1$.

$$A + 1 = (nA - 200) / (n - 10) \rightarrow (A+1)(n-10) = nA - 200.$$

$$An - 10A + n - 10 = nA - 200 \rightarrow -10A + n - 10 = -200 \rightarrow 10A - n = 190. \dots(\text{Eq. 1})$$

Step 2: Formulate an equation for the second scenario

If the last 10 tests are not considered: Sum of last 10 tests = $10 \times 30 = 300$.

New total score $S'' = nA - 300$. New number of tests $n'' = n - 10$. New average $A'' = A - 1$.

$$A - 1 = (nA - 300) / (n - 10) \rightarrow (A-1)(n-10) = nA - 300.$$

$$An - 10A - n + 10 = nA - 300 \rightarrow -10A - n + 10 = -300 \rightarrow 10A + n = 310. \dots(\text{Eq. 2})$$

Step 3: Solve the system of linear equations

Add Eq. 1 and Eq. 2: $20A = 500 \rightarrow A = 25$.

Substitute $A = 25$ into Eq. 2: $10(25) + n = 310 \rightarrow n = 60$.

The total number of tests taken is 60.

Question 32

In a company, 20% of the employees work in the manufacturing department. If the total salary obtained by all the manufacturing employees is one-sixth of the total salary obtained by all the employees in the company, then the ratio of the average salary obtained by the manufacturing employees to the average salary obtained by the non-manufacturing employees is

A) 6:5

✓ B) 4:5 ← **Correct Answer**

C) 5:4

D) 5:6

Correct Answer: B

Solution:

Calculate the ratio:

$$\text{Ratio} = [S / (1.2E)] / [5S / (4.8E)]$$

$$= (S / 1.2E) \times (4.8E / 5S)$$

$$= 4.8 / (1.2 \times 5) = 4.8 / 6 = 4/5.$$

The ratio is 4:5.

